

Softwarica

COLLEGE OF IT & E-COMMERCE

SECURE PASSWORD GENERATOR



Aayush DC

ST4017CMD Introduction to Programming

Summited to: Mr Abishek Bimali

Student ID :250291

Contents

ABSTRACT.....	3
1.INTRODUCTION.....	4
2.PROBLEM STATEMENT.....	5
3.OBJECTIVE OF THIS PROJECT	6
4. LITERATURE REVIEW.....	6
5. SYSTEM REQUIREMENT	7
6. ALGORITHM DESIGN.....	7
7.DATA STRUCTURES USED	13
8. SYSTEM ARCHITECTURE AND INTERFACE.....	13
9. IMPLEMENTATION OF USING PYTHON.....	13
10. TESTING AND RESULTS	14
11. ETHICAL AND SECURITY CONSIDERATION.....	14
12. LIMITATION AND FUTURE ENHANCEMENTS.....	16
13.Uses of secure passwords by Different Sectors.....	16
14.Real-Life use of Secure Passwords	17
15.Real life incidents caused by weak password	18
16. Levels of Secure Passwords.....	20
17.The main purpose of a secure password generator project	24
18. CONCLUSION	25

Acknowledgements

I would like to express my heartfelt thanks to my course teacher, Mr. Abhishek Bimali, for his constant guidance and support during the completion of this coursework on "Development of Secure Password Generator Using Python." With his teaching and guidance, I have been able to understand the various important concepts associated with cybersecurity and password generation in a better and more effective manner.

I would also like to express my gratitude to Softwarica College for offering a suitable academic environment, facilities, and resources, which have enabled this coursework to be completed. The material provided was greatly helpful in assisting a better understanding of the subject matter.

Finally, I would like to thank my friends and classmates for their cooperation, discussions, and motivation during the project work. This coursework has helped me improve my knowledge of Python programming and its real-world significance with respect to the use of a good and secure password.

- Aayush DC

ABSTRACT

Passwords are the initial defense mechanism in contemporary cybersecurity environments. However, weak and reused passwords continue to be one of the top reasons for data breaches and unauthorized access. This project discusses the design and development of a secure password generator using Python, which is intended to generate highly secure and unpredictable passwords based on the user's defined security needs. The proposed system employs algorithmic approaches such as randomization, conditional statements, and customized data structures to increase the complexity and unpredictability of the generated passwords. A command-line interface has been developed to enable user interaction. The project also addresses the ethical and security implications of using the developed system responsibly.

1)INTRODUCTION

In the current digital age, passwords are an essential component in the protection of confidential data on computer systems, networks, and online applications. As the main form of authentication, passwords are used in online applications such as email and banking systems. However, many people continue to use weak passwords that are easily guessed, such as common words, personal information, and repeated characters. Such passwords make computer systems vulnerable to cyber attackers such as brute-force attacks, credential stuffing, and dictionary attacks.

Cybersecurity experts have highlighted the need for strong password policies to counter such threats. A strong password should consist of a combination of uppercase letters, lowercase letters, numbers, and special characters, making it difficult for hackers to guess or crack. Despite such knowledge, the creation and recall of strong passwords continue to pose a challenge to users.

The proposed project involves the development of a secure password generator using Python programming. The project will enable the automation of the generation of strong passwords through algorithmic logic, with the capacity for users to set security parameters. The project is relevant to cybersecurity concepts and the application of algorithms, data structures, and secure coding techniques.

2)PROBLEM STATEMENT

In the modern technology era, passwords have become the most common form of authentication used to secure personal accounts, business systems, financial information, and sensitive data. Nevertheless, many people continue to use weak passwords like simple words, names, birthdays, and generic patterns. Such weak passwords make systems extremely susceptible to cyberattacks like brute-force attacks, dictionary attacks, and credential stuffing.

There have been many real-life examples that have revealed how weak or stolen passwords can cause massive data breaches. For instance, the LinkedIn data breach revealed the passwords of millions of users because of weak password protection strategies, and the Equifax data breach in 2017 proved that poor cybersecurity measures can cause significant financial and reputational loss.

Despite awareness and cybersecurity best practices, users still face challenges in creating secure passwords that are complex and unique. Some online services offer password suggestions, but users tend to disregard them because of the complexity of remembering them. Moreover, some simple password generators have simple algorithms and lack randomness, making them less secure.

Thus, the main problem that this project aims to solve is:

How can we develop and create a secure password generator that can produce strong, random, and unpredictable passwords while being user-friendly and following cybersecurity best practices?

The system should:

- Use a secure random password generation algorithm.
- Use a combination of uppercase letters, lowercase letters, numbers, and special characters.
- Be customizable for password length.
- Have high entropy to withstand brute-force and dictionary attacks.
- Follow current cybersecurity best practices.

By solving this problem, the project will help decrease the vulnerability of unauthorized access, improve user awareness of security, and promote safe online practices in both personal and business settings.

3.OBJECTIVE OF THIS PROJECT

The primary goals of this project are:

1. Designing an algorithm to produce secure and random passwords
2. Developing the algorithm using Python
3. Developing a command-line interface for interaction with the user
4. Using custom data structures rather than just relying on built-in data structures
5. Showing how cybersecurity best practices can be followed while designing passwords
6. Using the tool in an ethical manner

4. LITERATURE REVIEW

Password security is an area that has been extensively researched in the realm of cybersecurity. It has been found that weak passwords are one of the most commonly exploited vulnerabilities in password security systems. As per cybersecurity reports, a large percentage of data breaches are caused by compromised credentials.

The conventional password system has traditionally included complexity requirements. However, users have been observed to reuse passwords on various platforms. Password managers and generators have been developed to counter this problem. However, users are often skeptical about using third-party software or lack knowledge about how these systems function.

Python has emerged as a preferred programming language for various cybersecurity applications because of its ease of use, flexibility, and rich libraries. Algorithmic password generators use the principles of randomness and entropy to generate unguessable passwords. It has been stressed in various studies that algorithmic password generation systems can drastically lower the success rate of brute-force attacks.

The proposed project aims to develop a transparent, customizable, and ethical password generation system based on these principles.

5. SYSTEM REQUIREMENT

5.1 Functional Requirement

1. User can choose password length
2. User can select character types (letters, numbers, symbols)
3. System generates a secure password
4. Password strength is displayed

5.2 Non-Functional Requirement

1. The system should be easy to use
2. Passwords should be unpredictable
3. The application should run efficiently
4. Ethical use must be ensured

6. ALGORITHM DESIGN

The password generation algorithm uses a structured approach:

1. Accept user input for password length
2. Ask the user to choose character sets
3. Store the chosen characters in a custom data structure
4. Randomly choose characters based on length
5. Combine characters to generate a password
6. Display the generated password

Pseudocode:

```
import secrets  Untitled-1  import secrets  Untitled-2  aayush.py X
C: > Users > aayus > aayush.py > ...
1
2  import secrets
3  import string
4
5  def generate_secure_password(length):
6      if length < 8:
7          print("Password length must be at least 8 characters")
8          return
9
10     lowercase = string.ascii_lowercase
11     uppercase = string.ascii_uppercase
12     digits = string.digits
13     symbols = string.punctuation
14
15     password = [
16         secrets.choice(lowercase),
17         secrets.choice(uppercase),
18         secrets.choice(digits),
19         secrets.choice(symbols)
20     ]
21
22     all_chars = lowercase + uppercase + digits + symbols
23     for _ in range(length - 4):
24         password.append(secrets.choice(all_chars))
25
26     secrets.SystemRandom().shuffle(password)
27
28     print("Generated Secure Password:", "".join(password))
29
30
31  # ---- Main Program ----
32  length = int(input("Enter password length: "))
33  generate_secure_password(length)
34
```

This algorithm ensures randomness and flexibility while maintaining simplicity .

OUTPUT

```
Enter desired password length: 20
Your Secure Password: A{Dl~OtssuS49[S,I.OD
```

(Secure password generator - Code Explanation)

1.Importing Required Modules

```
import secrets
import string
```

- **Secrets**

1. It is used for cryptographically secure random numbers
2. It is much safer than random
3. It is used in real security systems likes (banks, passwords)

- **String**

It provides ready-made characters sets like :

1. Symbols
2. Digits
3. Uppercase letters
4. Lowercase

2. Defining the Functions

```
def generate_secure_password(length):
```

- It helps to creates a function named called (generate_ secure_ password)
- Length=It is a user input how long the password should be

3.Minimum Length check

```
if length < 8:  
    print("Password length must be at least 8 characters")  
    return
```

- As we can see in the input 8 is greater than length and if the user enters a password length less than 8 :
 1. Program will show a warning
 2. And stop execution (return)

4.Character Sets

```
lowercase = string.ascii_lowercase  
uppercase = string.ascii_uppercase  
digits = string.digits  
symbols = string.punctuation
```

The above variables stores following:

- Lowercase=a to z
 - Uppercase=A to Z
 - Digits=0-9
 - Symbols=* & ^ % \$ # @ ! etc
- As using all this combination of variables makes password strong as well as hard to guess

5.Ensuring password complexity

```
password = [  
    secrets.choice(lowercase),  
    secrets.choice(uppercase),  
    secrets.choice(digits),  
    secrets.choice(symbols)  
]
```

- The above following input helps to guarantees the password contains:

1. There should be at least one lowercase letter
2. There should be at least one uppercase letter
3. There should be at least one number
4. There should be at least one symbol

It helps to prevents from weak password like abcdefg123

6.Creating full character pool

```
all_chars = lowercase + uppercase + digits + symbols
```

- It helps to combine all character into one pool
- It is also used to fill the remaining password length

7.Filling Remaining Characters

```
for _ in range(length - 4):  
    password.append(secrets.choice(all_chars))
```

- It already used 4 characters (one from there each type)
- It helps to ass remaining characters randomly
- It uses secrets .choice () for security

8.Shuffling the passwords

```
secrets.SystemRandom().shuffle(password)
```

- It randomly rearranges the characters
- It helps to prevent fixed patterns like
(Lowercase→ Uppercase→ digits→ symbols)
It is also very important for the real world security

9.Printing the passwords

```
print("Generated Secure Password:", "".join(password))
```

- It helps to convert the password list into a string
- It Displays the final secure passwords

10.Main program (It is User input)

```
# ---- Main Program ----  
length = int(input("Enter password length: "))  
generate_secure_password(length)
```

- It takes password length from users
- It calls the password generator functions

7. DATA STRUCTURES USED

Instead of using Python's built-in list, a data structure was developed to hold character sets. This data structure enables better control over character selection and improves comprehension of algorithm design.

The use of custom data structures shows a better understanding of programming concepts and meets the module requirements.

8. SYSTEM ARCHITECTURE AND INTERFACE

The system is designed using a command-line interface (CLI). The CLI enables users to interact with the application using text-based menus. The CLI is lightweight, efficient, and ideal for cybersecurity applications.

1. The system architecture comprises:
2. Input handling module
3. Password generation module
4. Output display module

9. IMPLEMENTATION OF USING PYTHON

The project is developed using the Python programming language. The random module in Python is used for secure character selection. Conditional statements and loops are the basic logic of the algorithm.

Error handling is also added to avoid invalid inputs such as negative password length. The code is managed using GitHub Classroom.

10. TESTING AND RESULTS

The system was tested for various password lengths and combinations of characters. The test results revealed that the passwords were strong and varied, thus difficult to predict.

1. The test cases included:
2. Minimum password length
3. Maximum password length
4. Use of all character types

The system worked well under all test conditions.

11. ETHICAL AND SECURITY CONSIDERATION

When creating a secure password generator, it's essential to consider both its technical functionality and the ethical and security obligations associated with it. A password generator is developed to enhance security; however, if not properly designed, it may introduce new vulnerabilities. Consequently, both ethical and security issues need to be regarded with importance.

Initially, from a security perspective, the key element is randomness. A reliable password generator must employ a robust and random technique to generate passwords. Using a basic or weak random function might allow attackers to guess or foresee the password sequence. That undermines the entire aim of the tool. The generator should depend on secure algorithms that create passwords that are genuinely random and hard to break

Password strength is something you really need to think about. The password generator needs to make passwords that're long and have a mix of uppercase letters, lowercase letters, numbers and special symbols. If passwords are short or simple they are easy to guess. People can get into your account. If the tool makes passwords people might think they are safe but they are not. This makes people feel safe when they are not. That is very bad.

You also need to think about privacy. A password generator should never. Write down the passwords it makes. It should not send the passwords it makes to any computer. People trust the tool to help them keep their accounts safe. If the tool secretly keeps passwords it is not doing what people trust it to do. That can cause big problems with peoples information. The people who make the tool have a responsibility to keep peoples information private.

Transparency is really important when it comes to these tools. People who use the tool need to know how the tool works. If the tool says it is secure then it should tell people how it keeps them safe. It is not right to hide information or make false claims about security. When the people who make the tool are honest about how it works that is when people start to trust them.

The people who make the tool also need to follow the rules for writing code. If the tool has weaknesses bad people might find a way to use them. The people who make the tool should check what people put into it not use codes and update the tool all the time. A tool that is supposed to keep people needs to be safe itself or it will actually make things worse.

There are also laws that the people who make the tool need to think about. In a lot of places there are rules, about how to handle the information that people give them. Even if the tool does not ask for information it should still follow the rules and be safe. If the people who make the tool do not do what the law says they could get in trouble.

Another thing to think about is using the tool in a way. Even though a password generator is meant to keep people safe some people might use it for things. The people who make the tool cannot control everything people do with it but they should say clearly that the tool is only meant to be used for legal things. The people who make the tool should also help people understand how to keep their information safe on the internet. Password generators, like this one need to be used in a way to keep everyone safe.

So when we make a password generator it is not, about making something that gives us random characters. We have to be responsible and honest and know about the things that can happen. The people who make these things have to make sure the passwords are really random keep the users information do things in a safe way and help people use them in a good way. Security and ethics are very important. They go together. If we do not pay attention to one of them the whole thing does not work well. Password generators need to be secure. The people who make them need to think about ethics.

12. LIMITATION AND FUTURE ENHANCEMENTS

Limitations:

- 1) It is CLI-only interface
- 2) There is no password storage functionality

Future Enhancement:

- 1) It is GUI-based interface
- 2) Integration with password managers
- 3) Password strength visualization
- 4) Multi-factor authentication supports

13. Uses of secure passwords by Different Sectors

As we know secure password plays a vital role in protecting digital systems from unauthorized access. In today's real world, especially in businesses and banks, passwords are the first line of defense against cybercrime like fraud, data theft and hacking.

Use of secure passwords in Banking systems

Banks also ensure the security of customers' accounts through the use of strong passwords in online or mobile banking. Once an individual logs in to their online account, the strong password ensures that only the actual customer can access their information. This prevents hackers from accessing the customers' money.

Use of secure password in Business and Organizations

In businesses, passwords play an important role in ensuring the security of business internal systems, such as the employee interface, email services, as well as employee payrolls and other company data. The main aim of using secure passwords is to ensure that only authorized employees have access to such data.

Use of secure password in Personal Digital Accounts

Secure passwords are commonly used to secure personal accounts like email accounts, social media accounts, cloud storage, and online shopping sites. As many of these services are also associated with email accounts, a secure password also prevents identity theft and misuse.

Use of secure password in Government and Educational system

Government portals and educational sites use strong passwords to keep citizens' records, students' data, and examination results confidential and protect these from any kind of misuse.

Use of secure password in Remote work and cloud service

As remote work increases, secure passwords protect access to cloud platforms and virtual private networks and remote desktops. This allows employees to work from anywhere without company systems being exposed to possible cyber threats.

14.Real-Life use of Secure Passwords

The need for secure passwords in the protection of digital information is a common phenomenon in real-life situations. The use of computers and the internet is a common phenomenon in the lives of individuals, businesses, and governments.

For instance, passwords are used to secure Internet banking accounts and mobile banking accounts by banking institutions. With secure passwords, bank customers are able to access their accounts and perform various activities such as checking their accounts and sending bills. This helps to ensure that only the owner of the bank accounts can access the information.

In business organizations, passwords are used to regulate and protect company systems such as employee portals, email accounts, databases, and payroll systems. Every employee is allocated a robust and unique password for the purpose of maintaining confidentiality of the company's vital information, customer information, and business strategies.

In one's life, secure and unique passwords are crucial to securing email accounts, social media sites, cloud storage services, and shopping sites. Used email accounts can act as a gateway to numerous other accounts, and thus, a strong email account password can assist one in preventing identity theft and protecting one against unauthorized account recovery.

The importance of strong passwords is also necessary for both the government and educational systems. Government websites provide strong passwords for citizen information, which are important for national ID cards, citizen taxes, and citizen service accounts. Similarly, educational systems provide strong passwords for students to get information on their examination results and learning materials.

With the increase in people working remotely and cloud computing, secure passwords provide access to cloud services, virtual private networks, and remote desktops, hence providing security for an organization despite the number of people working remotely.

15. Real life incidents caused by weak password

1. LinkedIn Password Leak (2012)

In 2012, the online professional network site LinkedIn was the victim of a massive cyber-attack where the passwords to the accounts of its millions of users were stolen and made publicly available online.

- The hackers gained access to the database and released hashed passwords, which were weak and not adequately protected, and therefore easily cracked.

Boarder impacts

1. In 2016, an expanded set of data from the same breach, including over 100 million email addresses and password hashes, was posted online, demonstrating the continued risk years later.
2. The response by LinkedIn was to make many of its users reset their passwords and work on security practices.

2.Yahoo Voice Password Leak (2012)

About 450,000 Yahoo Voice usernames were exposed due to the hacking of the website. The passwords were not encrypted, meaning that they were available to the attackers, resulting in a huge loss.

Key facts

1. The compromised passwords are kept in plaintext, i.e., they are not encrypted, and thus no decryption is required by the hacker to obtain the password.
2. As users continued to use the same password elsewhere, the risk extended beyond Yahoo Voice.

(LinkedIn Long-Term Risk from Weak Passwords (2016 Revelation))

As mentioned earlier, the LinkedIn data breach took place in 2012, but in 2016, the dimensions of the breach were revealed. The breach, which was originally described as a leakage of 6.5 million hashed passwords, was revealed to be a leakage of email addresses and hashed password data of over 100 million users, all of which were from the 2012 breach. LinkedIn noted that many users who were breached in 2012 did not change their passwords, thus remaining vulnerable even after all those years.

Boarder Impact

1. Users: Accounts were still vulnerable, and risks of identity theft, phishing, and hacking of other services persisted.
2. LinkedIn: Reputation damage, loss of user trust, forced password reset, and enhanced security policy.
3. Cybersecurity Industry: Emphasized the need for secure password storage, 2FA, and response to breaches.
4. Society: Raised awareness about using strong and unique passwords, along with password reuse avoidance.

16. Levels of Secure Passwords

There are 5 levels of secure password they are listed below:

Level 1: Very weak Password (which is easy to hack)

It's characteristics are:

1. It is short less than 6 characters
2. It has only letter or number
3. It has Common words or patterns

It's Examples:

- 123456
- Password
- Qwerty

It is easy to hack Because of following reasons:

1. It can be cracked in seconds using dictionary attacks
2. It has very common password

Level 2 : Weak Password

It's characteristics are :

1. It is slightly longer than 6-7 characters
2. It is combination of both letters and numbers
3. It is in predictable pattern

It's Example:

1. Aayush123
2. Hello2024
3. Admin12

It is easy to hack Because of following reasons :

1. Someone's personal information is easy to guess
2. It has Low security
3. It is vulnerable to brute-force attacks

Level 3 :Moderate Password

It's characteristics are listed below:

1. It has 8 to 9 characters
2. It has the combination of uppercase + lowercase +numbers
3. It has no symbols

It's example are :

1. Cyber123
2. Bankuser9
3. Secure98

According to Security level

1. It is Acceptable for normal websites
2. It is still easy to hack with time

Level 4: Strong Password (Hard to Hack)

It's characteristics are listed bellow :

1. It has 10 to 12 characters
2. It has the combination of uppercase+ lowercase+ numbers + symbols
3. It has random letters no personal information

It's Examples are listed below:

1. cyb@r9!Qe2
2. Mon#s3cU!7

It is hard to hack because of following reasons :

1. It has the resistant to brute-force attacks
2. It is hard for human and bots to guess and has no personal information
3. It is highly recommended for business

Level 5 : very strong Password (Extremely Hard to Hack)

IT's characteristics are listed below :

1. It has the combination of more than 14+ characters
2. It is fully random hard to guess
3. It uses all character types
4. It is often system-generated

It's example are listed below :

1. T9\$e!Qp7@Z#2Lm
2. A7#xP!9@Qm2\$K

It is extremely secure because of following reasons

1. It takes years or centuries to crack
2. It is Used with 2FA/OTP

Password security level Table

Level	Password Types	Hack Difficulty	Sector used
Level 1	Very weak	Very Easy	We should Never used
Level 2	weak	Easy	We should avoid this passwords
Level 3	Moderate	Medium	Used in normal websites
Level 4	Strong	Hard	Used in Businesses
Level 5	Very strong	Extremely Hard	Used in Banks & finance

The strength of password security varies from weak passwords, which can easily be cracked using basic hacking techniques, to strong passwords, which are combinations of long random characters, hence difficult to crack even with the most advanced hacking tools.

17.The main purpose of a secure password generator project

Mainly, a password generator project aims to automatically create strong, random, and unpredictable passwords that protect user accounts and systems from unauthorized access. The project is designed to provide better security rather than relying on human-chosen passwords, usually weak or reused, by generating passwords using secure algorithms.

This project provides avoidance of users' mistakes concerning password planning, such as using simple passwords, personal information, or identical ones on different platforms. By mixing uppercase and lowercase letters with numbers and special symbols, generated passwords could become too difficult to either guess or crack with the methods of brute-force or dictionary attacks.

In real life, it finds its usefulness for generating secure passwords within banking systems, business organizations, and cloud services, even with one's personal account, considering strong authentication. This helps ensure good practice in cybersecurity by reducing or avoiding data breaches, identity theft, and financial loss.

This project also shows various cybersecurity concepts in practice, like the use of cryptographically secure randomness, control of access, and the treatment given to password policies, which make it very useful both for academic learning and real application.

18. CONCLUSION

In conclusion, the project on Secure Password Generator successfully proves the need for adherence to good password management habits in today's digital world. As cyber attacks escalate in this digital era, insecure and easily predictable passwords are perhaps the most common cause of security threats. This project removes that hindrance by providing a reliable means of automating password creation.

The concept of the project can be helpful in eliminating the need for users to create their own passwords, which can often be weak and involve simple words, personal information, and repeated passwords for various systems, as is done presently. With the generation of new passwords containing a mix of uppercase, lowercase, numerical, and symbol characters, a high level of security can be guaranteed. The implementation of cryptographically strong random functions can also provide a high level of security, as guessing such passwords is extremely hard.

From a technical standpoint, the applications of this project can be seen in various areas such as the banking sector, business organizations, education sector, and digital account information of people. This helps protect our sensitive data from unauthorized use. By adopting the use of passwords in the right manner, this project contributes to the reduction of cybercrime activities occurring in the country.

Besides the practical advantages, the Secure Password Generator project can foster knowledge of basic security concepts. One can get hands-on experience with concepts like secure coding, access control, and password techniques. This makes this project useful not only as a security solution but also as a learning experience for students and newbies in security matters.

In summary, this project underscores the fact that passwords and their security cannot be taken for granted and are instead integral components of network security. The Secure Password Generator provides an effective tool for securing passwords, engaging users in secure internet practice, and enhancing internet security as a whole.

References

NIST Special Publication 800-63B.

Authentication cheat sheet¶ (no date) *Authentication - OWASP Cheat Sheet Series.*

Alextjokro (2025) *The importance of password security in the Digital World - Robere & Associates Indonesia, Robere & Associates Indonesia*

Password strength (2026) *Wikipedia.* Available at: https://en.wikipedia.org/wiki/Password_strength#:~:text=The%20strength%20of%20a%20password,the%20strength%20a%20large%20amount. (Accessed: 07 February 2026).

Random password generator (2025) *Wikipedia.* Available at: https://en.wikipedia.org/wiki/Random_password_generator (Accessed: 07 February 2026).

2012 yahoo voices hack (2025) *Wikipedia.* Available at: https://en.wikipedia.org/wiki/2012_Yahoo_Voices_hack (Accessed: 07 February 2026).

Allah, M.R. (2026) *16B passwords leaked: Password-based security is broken: Mohamed Rida Allah posted on The topic, LinkedIn.* Available at: https://www.linkedin.com/posts/iamrida_the-day-16-billion-passwords-died-and-why-activity-7422745478682968064-uk5 (Accessed: 07 February 2026).

Cybersecurity 101: Why choosing a secure password is so important (no date) *Walden University.* Available at: <https://www.waldenu.edu/programs/information-technology/resource/cybersecurity-101-why-choosing-a-secure-password-in-so-important> (Accessed: 07 February 2026).

Alextjokro (2025) *The importance of password security in the digital world.* Robere & Associates Indonesia. Available at: <https://robere.co.id/the-importance-of-password-security/> (Accessed: 7 February 2026)

Bonneau, J. (2012) 'The science of guessing: Analyzing an anonymized corpus of 70 million passwords', *Proceedings of the IEEE Symposium on Security and*

Privacy. Available at: <https://ieeexplore.ieee.org/document/6234435> (Accessed: 24 February 2026).

Wikipedia (2025) *Random password generator*. Available at: https://en.wikipedia.org/wiki/Random_password_generator (Accessed: 7 February 2026).

Microsoft (2023) *Password guidance and best practices*. Available at: <https://learn.microsoft.com/en-us/security> (Accessed: 24 February 2026).